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Conversational Counterfactual Guidance for Operator-Centred Hyperparameter Optimisation in Industrial Anomaly Detection

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Abstract:	Hyperparameter optimisation is essential for high-performing machine-learning systems, yet its logs are rarely reused for interpretation. In industrial visual anomaly detection, configuration must balance detection quality against training-data availability, review effort, and data quality. Optimisation tools expose scores or rankings, but rarely show how a configuration should change to satisfy operator needs. We present Counterfactual Explainable Hyperparameter Optimisation (CF-XHPO), which reuses trials to train a surrogate and generate constrained what-if recommendations through a schema-constrained conversational interface. We separate surrogate-level query satisfaction from main-model confirmation by retraining PatchCore for 24 recommendations each on the MVTec Anomaly Detection benchmark and the Proditec production dataset. All 48 satisfy their surrogate queries. On Proditec, 21/24 meet the target after retraining, with mean confirmed F1 score of 0.815, mean surrogate-to-model gap of −0.024, and test area under the receiver operating characteristic curve (AUROC) of 0.947. MVTec exposes a reliability boundary: despite a grouped cross-validated coefficient of determination ($R^2 = 0.995$), none of the 24 reaches the target. The discrepancy occurs on rerun validation data (−0.204), while validation-to-test transfer is small (+0.013); models retain per-category AUROC of 0.827 although pooled AUROC falls to 0.497. These findings motivate query-region evaluation, category-specific calibration, and main-model confirmation. On these logs, constrained retrieval matches off-log generation, supporting a retrieve-first policy when coverage is adequate. A 22-participant study finds high task validity in both workflows, with conversational participants reporting greater ease of option comparison. Parser evaluation identifies deterministic handling of relative constraints as a key requirement.

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To the Editors of *Future Generation Computer Systems*,

Please find enclosed our manuscript, *"Conversational Counterfactual Guidance for Operator-Centred Hyperparameter Optimisation in Industrial Anomaly Detection"*, submitted for consideration in the special issue *"Powering Data-driven Innovation: Responsible, Sustainable, Scalable, and Trustworthy AI"*.

Prior conference publication. This manuscript extends our workshop paper, S. Manai, S. Bobek, M. Asadi, A. Nowé, G. J. Nalepa, *"Counterfactual User Guidance for Improving Transparent Hyperparameter Tuning"*, submitted to and accepted at SmartSys'26, a workshop of the International Conference on Computational Science (ICCS 2026), held in Hamburg, Germany, 29 June – 1 July 2026, and published in *Computational Science – ICCS 2026 Workshops*, LNCS vol. 16789, Springer. The original paper is cited in the submission, the present title differs from it, and Section 2.6 of the manuscript is devoted to the comparison.

What carries over, and what does not. Two ideas are retained: the separation of hard model hyperparameters from soft, human-controllable parameters, and the reuse of a completed optimisation log as the basis for counterfactual configuration queries answered through a surrogate. Beyond these, the two papers share no datasets, no main model, no evaluation protocol, and no interface.

The conference study ran on four public tabular binary-classification benchmarks with XGBoost. Soft parameters there described data-handling actions — training-row fraction, injected missingness, resampling, feature fraction, labelled ratio — which an operator fixed manually before DiCE and CFNOW varied the model hyperparameters around a single factual seed. The reported evidence was an aggregate per-dataset comparison of surrogate, generated, and retrained F1 over five seeds, plus one case study showing a moderate surrogate-to-model correlation (Pearson $r = 0.53$, 16 candidates). That paper contained no structured query schema, no natural-language interface, no evaluation with human participants, and no industrial use case; a language-model layer and the Proditec deployment appeared only as announced future directions. The new material therefore stands well above the 40% threshold.

New contributions and where they appear.

- **A new application domain and main model** (Sections 3.1, 4.1–4.2). The framework is rebuilt for industrial visual anomaly detection over a PatchCore pipeline, evaluated on the MVTec AD benchmark and on the proprietary Proditec production line. The soft controls are redefined as deployment-facing quantities — training fraction, review budget, and assumed corruption level — with the review budget tied to an explicit operator thresholding rule.
- **Main-model confirmation as the reporting standard** (Sections 4.4–4.6). Every recommendation is retrained with the underlying model: 24 candidates per dataset plus 12 further confirmations in Section 4.9, for 60 confirmed configurations, reported per candidate rather than in aggregate.
- **Score-level diagnosis of the confirmation gap** (Section 4.7). An analysis of raw per-image anomaly scores localises the MVTec gap to the validation side (-0.204 prediction-to-rerun-validation against $+0.013$ validation-to-test transfer) and separates it from a distinct deployment problem: incommensurable per-category score scales reduce pooled AUROC from 0.827 to 0.497, and per-category thresholding recovers $+0.248$ mean F1 without changing the models. This yields the notion of *selection-conditioned fidelity* and a concrete calibration recommendation.
- **A validated query schema and conversational interface** (Sections 3.5–3.6). Manually fixed soft parameters are replaced by three objective types with free, fixed, and range constraints, plus deterministic

validation, repair, snapping, overlay, and an explicit fallback, together with automatic factual-anchor resolution.

- **Constrained retrieval as a first-class alternative to generation** (Section 4.9). A twelve-query suite at three anchors per dataset, with 12 additional confirmations, supports a retrieve-first policy under adequate log coverage.
- **Duplicate-leakage control and surrogate benchmarking** (Section 4.3): thirteen candidate regressors and grouped cross-validation by exact configuration signature, added because TPE revisits configurations and a shuffled split otherwise leaks them.
- **Anchor-sensitivity analysis** (Section 4.8): 80 additional surrogate-level candidates per dataset across five factual anchors.
- **Complete-interaction cost and reuse** (Section 4.10): component and end-to-end wall-clock measurement on a single power-capped workstation GPU.
- **Labelled evaluation of the conversational parser** (Section 4.11): 40 annotated operator utterances scored field-wise against two front-ends, isolating relative-constraint arithmetic as the failure mode.
- **Evaluation with human participants** (Section 5): a 22-participant between-subjects study with a post-task questionnaire and a qualitative analysis of written justifications.
- **Extended related work and limitations** (Sections 2 and 6.3), including the comparison subsection required by the journal's policy on conference extensions.

Note on double-blind review. The extension policy and the double-blind requirement interact: the manuscript must cite and compare against the earlier paper, which makes the authorship partly inferable. Following Elsevier's guidance, the prior work is cited and discussed in the third person. All other identifying material — author details, affiliation, funding, acknowledgements, and repository links — has been removed from the manuscript file and supplied separately.

Fit to the special issue.

- **Sustainable.** The framework reuses an optimisation log that has already been paid for. One surrogate prediction costs $\sim$2 ms against 31–67 s per PatchCore retraining, and the measured benefit at the level of a complete interaction is an amortised screening advantage of 1.02–1.74$\times$ (MVTec) and 2.12–3.80$\times$ (Proditec) over sequential retraining of the same eight alternatives. We report wall-clock cost only: no power telemetry was collected, so the paper makes no energy or CO_2 claim, and we state so explicitly rather than converting runtime into estimated savings.
- **Trustworthy and responsible.** The surrogate is a search layer, never evidence. Main-model confirmation is mandatory, the retrieval-based method is labelled honestly, and the manuscript reports a negative result and diagnoses it rather than presenting only the successful dataset.
- **Scalable.** The interactive layer runs on a single power-capped workstation GPU, the resource-constrained setting typical of factory deployments.
- **Data-driven innovation.** The pipeline extracts interactive value from data that every HPO run already produces, validated on a real industrial inspection use case alongside a public benchmark.

All authors have approved the manuscript and its submission. The manuscript is not under consideration elsewhere, and there are no competing interests to declare. The work was carried out within the Horizon Europe PEER project (GA No. 101120406). We suggest no reviewer exclusions.

Yours sincerely,

Sabri Manai
on behalf of the authors (Sabri Manai, Szymon Bobek, Grzegorz J. Nalepa)

Author Names, affiliations, and corresponding author details
(word or TEX file))

Conversational Counterfactual Guidance for Operator-Centred Hyperparameter Optimisation in Industrial Anomaly Detection

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CRediT authorship contribution statement

Sabri Manai: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data curation, Writing – original draft, Visualization. **Szymon Bobek:** Conceptualization, Methodology, Investigation, Data curation, Writing – original draft, Writing – review & editing, Supervision. **Grzegorz J. Nalepa:** Conceptualization, Writing – review & editing, Supervision, Resources, Funding acquisition.

Conflict of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Highlights

Conversational Counterfactual Guidance for Operator-Centred Hyperparameter Optimisation in Industrial Anomaly Detection

- CF-XHPO reuses HPO logs for constrained what-if configuration guidance
- On the production dataset, 21/24 recommendations confirm after retraining
- Global surrogate fidelity does not guarantee reliability in query-selected regions
- Constrained log retrieval matches off-log generation on dense HPO logs
- A 22-participant study finds greater ease of comparing options

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Abstract

Hyperparameter optimisation is essential for high-performing machine-learning systems, yet its logs are rarely reused for interpretation. In industrial visual anomaly detection, configuration must balance detection quality against training-data availability, review effort, and data quality. Optimisation tools expose scores or rankings, but rarely show how a configuration should change to satisfy operator needs. We present Counterfactual Explainable Hyperparameter Optimisation (CF-XHPO), which reuses trials to train a surrogate and generate constrained what-if recommendations through a schema-constrained conversational interface. We separate surrogate-level query satisfaction from main-model confirmation by retraining PatchCore for 24 recommendations each on the MVTec Anomaly Detection benchmark and the Proditec production dataset. All 48 satisfy their surrogate queries. On Proditec, 21/24 meet the target after retraining, with mean confirmed F1 score of 0.815, mean surrogate-to-model gap of −0.024, and test area under the receiver operating characteristic curve (AUROC) of 0.947. MVTec exposes a reliability boundary: despite a grouped cross-validated coefficient of determination ($R^2 = 0.995$), none of the 24 reaches the target. The discrepancy occurs on rerun validation data (−0.204), while validation-to-test transfer is small (+0.013); models retain per-category AUROC of 0.827 although pooled AUROC falls to 0.497. These findings motivate query-region evaluation, category-specific calibration, and main-model confirmation. On these logs, constrained retrieval matches off-log generation, supporting a retrieve-first policy when coverage is adequate. A 22-participant study finds high task validity in both workflows, with conversational participants reporting greater ease of option comparison. Parser evaluation identifies deterministic handling of relative constraints as a key requirement.

Keywords: hyperparameter optimisation, counterfactual explanations, anomaly detection, conversational interface, explainable artificial intelligence, industrial decision support

1. Introduction

Industrial visual inspection increasingly relies on machine-learning models to detect defects, reduce manual inspection effort, and support more consistent quality-control decisions. Deep learning has driven rapid progress in this area [1], with benchmarks and methods such as the MVTec Anomaly Detection (MVTec AD) benchmark, PatchCore, and EfficientAD among the most widely used reference points [2, 3, 4]. In this setting, however, a useful model is not defined by detection performance alone. Engineers and operators must also consider practical constraints such as the amount of training data available, the expected review effort, the quality of the input data, and the cost of retraining or deploying a new configuration. Hyperparameter optimisation (HPO) is therefore not only a technical search problem, but also a decision-support problem in which different configurations represent different operational trade-offs.

Automated HPO frameworks are effective at exploring large configuration spaces and identifying high-performing trials, as illustrated by tools such as Optuna and broader work on HPO benchmarking and multi-fidelity evaluation [5, 6, 7]. Yet, their outputs are usually presented as ranked configurations, optimisation traces, visual analytics views, or parameter-importance summaries [8, 9, 10]. These outputs help users understand which configurations performed well, but they provide limited support for actionable what-if questions: what should change if the operator wants similar quality with less review effort, a lower training fraction, or better robustness to data corruption? This limitation is particularly relevant in industrial contexts, where users often reason in terms of constraints and preferences rather than low-level model hyperparameters.

This paper addresses this problem by proposing Counterfactual Explainable Hyperparameter Optimisation (CF-XHPO), a framework that turns HPO logs into an interactive recommendation layer. Instead of discarding optimisation trials once the best configuration is found, CF-XHPO uses the log to train a surrogate of the configuration–performance relationship and to retrieve or generate alternatives under new operator constraints. The surrogate provides a millisecond-scale screening layer; main-model confirmation remains the final evidence for a selected configuration. On top of this layer, the framework compares complementary counterfactual-generation methods with constrained log retrieval, building on work on diverse, feasible, and actionable counterfactual explanations [11, 12, 13, 14]. These alternatives respect a separation between hard model hyperparameters and soft operator-facing controls, so recommendations stay technically valid while remaining expressed in terms operators can act on. To support such interaction, CF-XHPO includes a schema-constrained conversational layer, motivated by recent work on natural-language model explanation and conversational explainable artificial intelligence (XAI) [15, 16, 17], through which operators can state goals in plain

language, compare ranked alternatives, and receive concise explanations of the main trade-offs.

We evaluate the framework on the MVTec visual anomaly-detection benchmark and the Proditec industrial inspection use case. Both use the same PatchCore pipeline and configuration space, and every primary recommendation is retrained with the underlying model. This design lets us distinguish whether a recommendation satisfies its structured query under the surrogate from whether its predicted quality survives main-model confirmation. We further examine the reliability of the query-selected region, compare off-log generation with constrained retrieval, measure complete-interaction cost, and conduct a user study of the conversational workflow.

Manai et al. [18] demonstrated soft-control-constrained counterfactual guidance on tabular classification benchmarks. The present paper extends that work, retaining its hard/soft control separation and log-reuse idea and rebuilding the remainder for industrial visual anomaly detection. Its main contributions are a two-dataset industrial evaluation with main-model confirmation of 60 configurations and score-level diagnosis of the 48 primary recommendations; evidence that global surrogate fidelity does not guarantee reliability in query-selected regions, including a selection-conditioned gap and pooled category score-scale failure; and a schema-constrained conversational layer with workflow evaluation covering factual-anchor sensitivity, constrained retrieval, multi-alternative runtime, a labelled parser test, and a 22-participant study (Sections 4.7–5). Section 2.6 sets out the comparison in full.

2. Related Work

The proposed framework sits at the intersection of four research threads: explainable HPO, counterfactual explanations, conversational XAI, and industrial anomaly detection. This section reviews each thread and explains how they combine in our work.

2.1. Explainable and Interactive Hyperparameter Optimisation

Hyperparameter optimisation (HPO) is central to modern machine-learning workflows. Frameworks such as Optuna provide flexible black-box search over large spaces while recording trial-level results [5], and surveys cover strategies from grid and random search to Bayesian, evolutionary, and learning-based approaches [19, 20]. Their outputs, however, are optimisation traces, best-trial summaries, or ranked configurations: dashboards and visual-analytics tools such as XAutoML [9] show *which* configurations performed well, not *how* a configuration should change to meet a new operator constraint, and surrogate-based importance analysis [21] serves benchmarking rather than user-facing recommendation.

A parallel effort makes HPO and automated machine learning (AutoML) more transparent: partial dependence plots explain hyperparameter effects across the explored space [8], ShrinkHPO exposes and prunes less promising search regions [10], explainable Bayesian optimisation targets users who need to understand the search process itself [22], and related work argues for opening optimisation procedures to inspection and human interpretation [23, 24].

Together, these works establish that transparency, interaction, and surrogate modelling matter in HPO and AutoML, but they mainly support optimisation analysis, benchmarking, or visual inspection rather than post-hoc configuration guidance under user-defined practical constraints.

Green artificial intelligence (Green AI) treats efficiency as an evaluation criterion and tracks model-development energy and carbon costs [25, 26, 27, 28]. Because HPO multiplies training costs, we report component and complete-interaction latency for log reuse; energy or carbon claims require power telemetry.

2.2. Counterfactual Explanations for Configuration Guidance

Counterfactual explanations describe how an input would need to change to obtain a desired model output [29]. They are widely studied in explainable artificial intelligence because they are contrastive, actionable, and often easier for users to interpret than global feature-importance summaries [30]. Diverse Counterfactual Explanations (DiCE) introduced a model-agnostic framework for generating diverse counterfactual explanations, emphasising multiple plausible alternatives rather than a single nearest change [11], and multi-objective methods show that validity, proximity, sparsity, and diversity can be treated as competing objectives [12]. Broader reviews organise this literature around validity, proximity, sparsity, diversity, feasibility, and actionability [30], distinguishing counterfactual explanations as target states from recourse as feasible interventions [31] and requiring that proposed changes be controllable and realistic for the user [32].

Feasibility has also been studied through data-supported counterfactual paths: Feasible and Actionable Counterfactual Explanations (FACE) argues that nearest counterfactuals may be unrealistic outside plausible regions of the data distribution and constructs paths supported by observed data [13], while CFNOW uses tabu search for model-agnostic counterfactual generation on tabular, image, and text data [14]. These approaches motivate treating retrieval from observed configurations and generation beyond them as empirical alternatives rather than assuming that off-data generation is always necessary. Spreitzer et al. show that formally valid counterfactuals are not automatically perceived as practical by users [33], an observation directly relevant to operator-facing HPO recommendations.

Counterfactuals have also been used to steer and personalise models interactively [34, 35, 36].

2.3. Conversational Interfaces for Human-Centred Machine-Learning Workflows

Conversational interfaces make machine-learning systems more accessible: TalkToModel shows natural-language dialogue helps users query and understand explanations without manually selecting tools [15], and recent work asks whether conversational XAI assistants improve understanding, trust, and

reliance over dashboards [17]. Both lower interaction barriers and warn that fluent explanations may increase overreliance where uncertainty is not communicated.

Design-oriented XAI research frames explanation needs as user questions that interfaces should support [16], and human-centred perspectives on explainability in the era of large language models emphasise users, contexts, and downstream effects over black-box opening [37, 38]. This perspective informs our user study, which evaluates the interface inside a decision-support workflow rather than in isolation.

In AutoML, human-centred studies show that users need support for understanding automation and deciding when to trust automated results [39, 40, 41], and interactive systems expose pipelines, optimisation histories, and trade-offs through inspectable interfaces [9, 42, 43, 44]. Our framework uses a schema-constrained conversational layer: the language model maps requests into validated objectives and constraints, while recommendation remains with the surrogate and counterfactual pipeline.

2.4. Industrial Visual Anomaly Detection as an Explanation-Critical Setting

Industrial visual anomaly detection aims to identify defective or unusual regions in product images, often under limited supervision. Benchmark datasets such as MVTec AD have become standard for evaluating unsupervised and semi-supervised anomaly-detection methods in industrial inspection scenarios [2]. Representative approaches span self-supervised, distribution-modelling, embedding-based, and deployment-oriented designs [45, 46, 3, 4].

Although anomaly-detection methods have become increasingly strong, deployment still depends on configuration choices such as backbone, image size, feature layers, coreset sampling, nearest-neighbour settings, and aggregation strategy. These technical choices interact with practical constraints such as available training data, review effort, latency, and tolerance to corrupted or noisy data. Standard benchmarks focus on detection metrics [2, 3], leaving less support for choosing among configurations with similar quality but different operational costs, as the Proditec production use case exemplifies.

2.5. Positioning of This Work

The reviewed threads provide the building blocks for this paper: HPO and AutoML automate configuration search, explainable HPO makes it inspectable, counterfactual explanations provide actionable alternatives, conversational XAI improves accessibility, and industrial anomaly detection motivates the deployment setting. Our work combines them by using HPO logs as the basis for constrained configuration recommendation exposed through a schema-constrained conversational interface. The evaluation separates three questions that are often conflated: whether a recommendation satisfies its query under the surrogate, whether the underlying model confirms the target, and whether users can act on the resulting alternatives. It also tests when retrieval from the existing log is sufficient and when generation beyond it adds measurable value.

2.6. Relation to the Conference Version

Manai et al. [18] introduced a preliminary version of this framework. Two ideas carry over from that work: the separation of hard model hyperparameters from soft, human-controllable parameters, and the reuse of a completed optimisation log as the basis for counterfactual configuration queries answered through a surrogate. The remainder has been rebuilt.

Their study was conducted on four tabular binary-classification benchmarks with XGBoost as the main model. Soft parameters there described data-handling actions such as the training-row fraction, injected missingness, resampling strategy, feature fraction, and the labelled ratio of a self-training scheme. An operator fixed those values manually, and DiCE and CFNOW then varied only the model hyperparameters around a single factual seed. Candidates were re-evaluated by retraining XGBoost over five seeds, and the reported evidence was an aggregate per-dataset comparison of surrogate, generated, and retrained F1, together with one case study showing a moderate surrogate-to-model correlation (Pearson $r = 0.53$ over 16 candidates). That paper contained no structured query schema, no natural-language interface, no evaluation with human participants, and no industrial use case; a language-model summarisation layer and the Proditec deployment were named only as future directions.

The present paper moves the framework to industrial visual anomaly detection with PatchCore, replaces manually fixed soft parameters with a validated query schema over deployment-facing controls, adds constrained log retrieval as a first-class alternative to generation, and makes main-model confirmation the reporting standard rather than an aggregate check. It also implements and evaluates the conversational layer that Manai et al. proposed, and evaluates the resulting workflow with human participants. Three supporting analyses are also new: a thirteen-model surrogate benchmark with grouped cross-validation and duplicate control (Section 4.3), an anchor-sensitivity sweep over five factual configurations (Section 4.8), and component and complete-interaction cost accounting (Section 4.10). Table 1 summarises the difference. The new material comprises the whole of Sections 4 and 5, the query schema and interface of Sections 3.5 and 3.6, and the problem formulation of Section 3.1.

3. The CF-XHPO Framework

This section presents the proposed framework for generating operator-centred counterfactual guidance from hyperparameter optimisation logs. The main idea is to transform optimisation trials into a reusable configuration–performance dataset, train a surrogate model over this dataset, and use the surrogate as a fast what-if layer for counterfactual recommendation. A schema-constrained conversational interface then allows users to express objectives and constraints in natural language, while the recommendation process remains grounded in validated structured queries, feature domains, and the candidate-production methods.

Table 1: Comparison with the conference version [18]. Dashes mark aspects not present in that paper.

Aspect	Conference version	This paper
Domain and main model	Tabular binary classification; XGBoost	Industrial visual anomaly detection; PatchCore (Sections 4.1–4.2)
Datasets	Four public tabular benchmarks	MVTec AD benchmark and the Proditec production line
Soft controls	Row fraction, injected missingness, resampling, feature fraction, labelled ratio; fixed manually per query	Training fraction, review budget, corruption level, tied to a deployment thresholding rule (Section 3.1)
Query specification	—	Validated schema with three objective types, repair, snapping, and explicit fallback (Sections 3.5–3.6)
Candidate production	DiCE and CFNOW	DiCE, ACFX, and constrained log retrieval, with a direct generation-versus-retrieval test (Section 4.9)
Confirmation evidence	Aggregate per-dataset comparison over five seeds	60 confirmed configurations with per-candidate gaps and target-satisfaction rates (Sections 4.4–4.6)
Reliability analysis	—	Score-level diagnosis, selection-conditioned fidelity, and threshold/score-scale decomposition (Section 4.7)
Natural-language layer	Named as future work	Implemented and evaluated on 40 labelled utterances across two front-ends (Section 4.11)
Human evaluation	—	22-participant between-subjects study with qualitative analysis (Section 5)

3.1. Configuration Guidance Problem

We consider an industrial visual anomaly-detection pipeline whose behaviour is determined by a configuration vector decomposed into two parts:

$$\mathbf{x} = (\mathbf{h}, \mathbf{s}), \tag{1}$$

where $\mathbf{h}$ denotes hard model hyperparameters, the technical choices internal to the anomaly-detection model, and $\mathbf{s}$ denotes soft operator-facing controls, the operationally meaningful choices an operator reasons about: the fraction of training data used, the review budget, and the assumed corruption level of the training data. The concrete variables instantiating $\mathbf{h}$ and $\mathbf{s}$ in our implementation are listed in Section 3.3 (Table 2).

The anomaly-detection model produces a continuous anomaly score a_i for each validation image i, rather than a binary decision. The review budget $\beta \in (0, 1]$ defines the fraction of highest-scoring images an operator inspects; this is a common deployment pattern in industrial inspection, where a human makes the final call on the images the model flags as most likely anomalous. Given n validation images, the number of reviewed images is:

$$m_\beta = \lceil \beta n \rceil. \tag{2}$$

The review threshold τ_β is the score of the m_β-th highest anomaly score. Each validation image also carries a ground-truth label $y_i^{\text{val}} \in \{0, 1\}$ indicating whether it is truly anomalous. The predicted anomaly label is:

$$\hat{y}_{i,\beta}^{\text{val}} = \begin{cases} 1, & a_i \geq \tau_\beta, \\ 0, & a_i < \tau_\beta. \end{cases} \tag{3}$$

The optimisation target for a configuration $\mathbf{x}$ is then:

$$v(\mathbf{x}) = F_1\left(\mathbf{y}^{\text{val}}, \hat{\mathbf{y}}_\beta^{\text{val}}\right). \tag{4}$$

The soft controls act at different points of the pipeline: training fraction sets how much normal data fits the model, corruption level sets the data-quality condition, and review budget sets the thresholding rule, an operator-facing deployment trade-off between detection quality and inspection effort rather than a model hyperparameter in the narrow sense.

The objective is not to find a high-performing configuration in isolation but to support guidance after optimisation has finished. The factual configuration $\mathbf{x}_0$, the operator's current starting point, is always instantiated as a logged trial: resolved automatically by the interface (Section 3.6) and fixed to specified trials for reproducibility in the technical evaluation (Section 4.2). Given the factual configuration $\mathbf{x}_0$, its surrogate-predicted quality $\hat{f}_\theta(\mathbf{x}_0)$, and a user request specifying a desired objective and soft constraints, the task is to return a set of alternative configurations:

$$C = \{\mathbf{x}'_1, \ldots, \mathbf{x}'_K\}, \tag{5}$$

where K is the requested number of alternatives. This turns hyperparameter configuration into a constrained counterfactual recommendation problem.

3.2. Framework Overview

Figure 1 illustrates the proposed framework. The process starts with an automated HPO run in which Optuna's tree-structured Parzen estimator (TPE) samples both hard hyperparameters and soft operator-facing controls. Each completed trial stores a configuration and its resulting validation F1 score, producing a trial-level configuration–performance log. This log is cleaned, typed, and transformed into a surrogate-ready dataset.

A surrogate model is trained to approximate the mapping from configurations to validation performance. The surrogate provides a reusable approximation layer that allows the system to explore local what-if alternatives without retraining PatchCore [3] for every user query. The candidate-production methods then search this surrogate space for configurations that satisfy a desired performance objective and selected soft constraints.

The final layer is the conversational query interface: natural-language goals are mapped to a structured query schema, validated, repaired, and snapped to the allowed domains before generation, and the output is a ranked list of configurations with predicted quality, soft-control values, constraint status, and textual trade-off explanations.

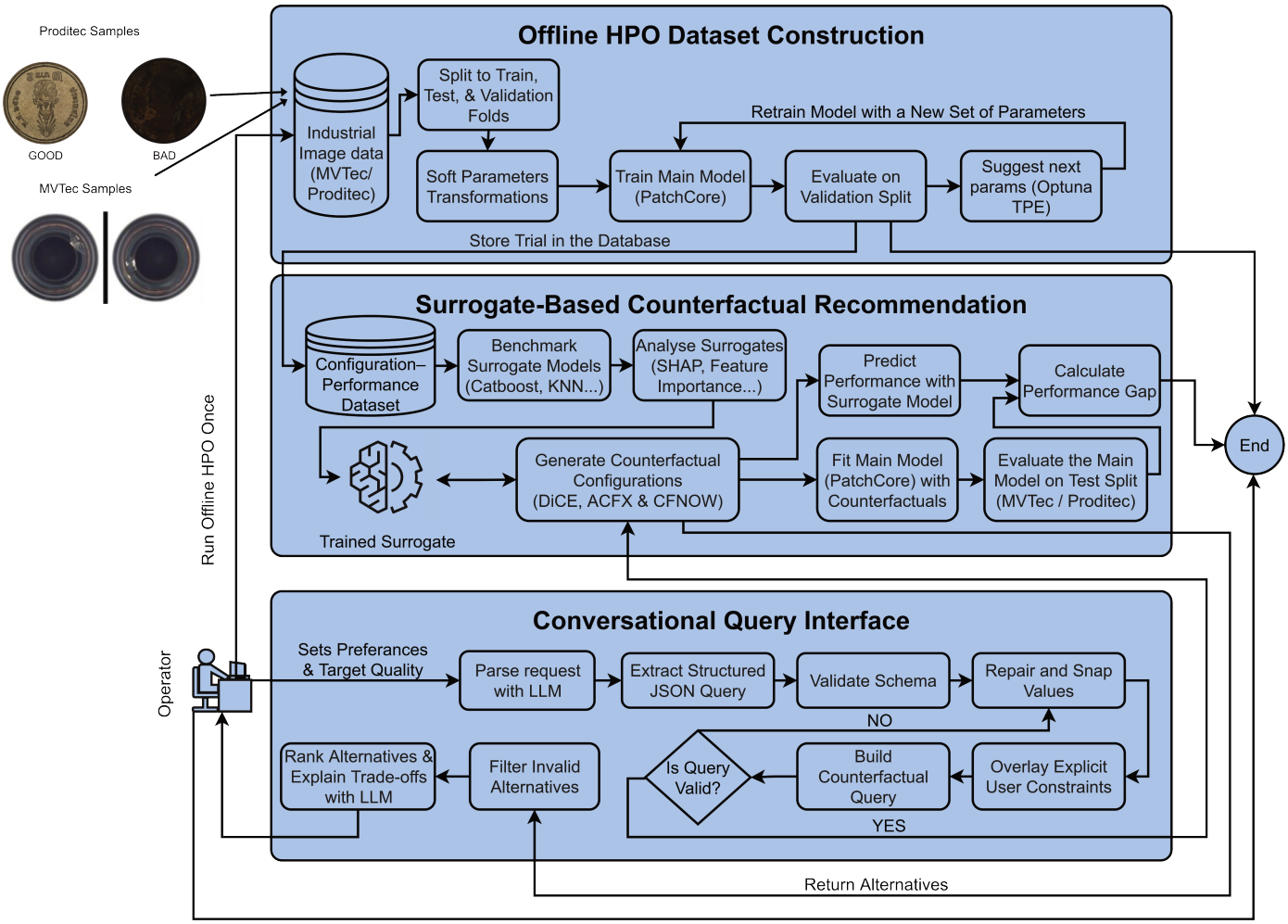


Figure 1: Framework overview. HPO denotes hyperparameter optimisation; TPE, tree-structured Parzen estimator; KNN, k-nearest neighbours; SHAP, Shapley additive explanations; DiCE, Diverse Counterfactual Explanations; ACFX, Actionable Counterfactual Explanations; LLM, large language model; and JSON, JavaScript Object Notation. CFNOW denotes the archived-code label for LogTopK in this figure, not the published CFNOW algorithm.

3.3. Optimisation Logs and Configuration Features

Each completed HPO trial produces a record:

$$(\mathbf{x}_i, v_i) = ((\mathbf{h}_i, \mathbf{s}_i), v_i), \tag{6}$$

where $v_i = v(\mathbf{x}_i)$ is the validation F1 score obtained by training and evaluating the pipeline under that configuration. Given N completed trials, the complete optimisation log is:

$$\mathcal{D}_{\log} = \{(\mathbf{x}_i, v_i)\}_{i=1}^{N}. \tag{7}$$

In our implementation, hard hyperparameters define the PatchCore-style pipeline: backbone, batch size, centre crop, coreset ratio, image size, feature layers, maximum patches per image, nearest neighbours, pretrained weights, and anomaly-score reduction; the soft controls are defined in Section 3.1. We deterministically cast categorical features, snap continuous controls, discard invalid or incomplete trials, and retain only features used by the surrogate and candidate-production methods. The cleaned MVTec and Proditec logs contain 9,976 and 2,999 trials, respectively.

Table 2 summarises the configuration variables used by the framework. The same configuration space applies to both the MVTec and Proditec use cases, enabling direct comparison of surrogate fidelity and counterfactual quality across the two datasets.

Table 2: Configuration variables used in the proposed framework (hard hyperparameters and soft controls).

Variable	Domain / Values
Hard hyperparameters	
Backbone	`wide_resnet50_2`, `resnet18`, `resnet34`, `resnet50`
Batch size	{8, 16}
Centre crop	{`none`, `0.875`}
Coreset ratio	[0.001, 0.101], step 0.002
Image size	{224, 256, 320}
Feature layers	{`l2`, `l3`, `l2_l3`, `l1_l2_l3`}
Maximum patches/image	{128, 256, 512, 1024}
Neighbours	[1, 30]
Pretrained weights	{`True`, `False`}
Reduction	{`max`, `mean`}
Soft controls	
Training fraction	[0.20, 1.00], step 0.01
Review budget	[0.005, 0.500], step 0.005
Corruption level	{`none`, `mild`, `strong`}

3.4. Surrogate-Based Performance Modelling

Training the main anomaly-detection model for every possible user query would be computationally expensive. The framework therefore trains a surrogate model:

$$\hat{f}_\theta : \mathcal{X} \to [0, 1], \tag{8}$$

where $\mathcal{X}$ is the mixed categorical–numerical configuration space and $\hat{f}_\theta(\mathbf{x})$ is the predicted validation F1 score. The surrogate is trained on a training split $\mathcal{D}_{\text{train}} \subseteq \mathcal{D}_{\log}$ (the exact split used in our experiments is given in Section 4.2) by minimising the squared prediction error:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{(\mathbf{x}_i, v_i) \in \mathcal{D}_{\text{train}}} \left(v_i - \hat{f}_\theta(\mathbf{x}_i) \right)^2 . \tag{9}$$

The trained surrogate serves two roles: fast approximation of model quality for unseen or modified configurations, and local what-if exploration that lets candidate-production methods search for configurations meeting a target without retraining PatchCore. Generated candidates need not coincide with any logged trial, and for such configurations the surrogate is the only source of estimates; whether generation beyond the log is *needed* is an empirical question of log coverage, tested directly in Section 4.9. Crucially, the surrogate is used as an interactive search layer, not as final evidence that a configuration will necessarily perform identically when retrained. Selected recommendations are therefore subject to main-model confirmation in the technical evaluation.

3.5. Counterfactual Configuration Generation

Given a factual configuration $\mathbf{x}_0$ and a user query, the counterfactual generation layer searches for alternative configurations $\mathbf{x}'$ that meet a desired performance objective while respecting user-defined constraints. The user objective is converted into a desired prediction interval $R(q, \mathbf{x}_0) = [L, U]$, where q denotes the structured query. The framework supports three objective types:

$$R(q, \mathbf{x}_0) = \begin{cases} [v, 1], & \text{if } q_{\text{obj}} = \texttt{target_min}(v), \\ [\hat{f}_\theta(\mathbf{x}_0) + \delta, 1], & \text{if } q_{\text{obj}} = \texttt{delta_improve}(\delta), \\ [L, U], & \text{if } q_{\text{obj}} = \texttt{target_band}(L, U). \end{cases} \tag{10}$$

Soft constraints restrict the candidate space for operator-facing controls. For numerical controls such as training fraction and review budget, a constraint can be free, fixed to a value, or restricted to a range. For the categorical corruption level, it can be free, fixed, or restricted to an allowed set. These constraints define a feasible region:

$$\mathcal{X}_q = \{\mathbf{x} \in \mathcal{X} : \mathbf{s} \text{ satisfies the soft constraints in } q\}. \tag{11}$$

A candidate satisfies its recorded query under the surrogate when:

$$\hat{f}_\theta(\mathbf{x}') \in R(q, \mathbf{x}_0), \qquad \mathbf{x}' \in \mathcal{X}_q. \tag{12}$$

We call the proportion of candidates satisfying Eq. 12 the *surrogate-level query satisfaction rate*. It answers whether candidate production and constraint handling return what was requested according to the model used for search. The *confirmation rate* instead measures the proportion whose retrained PatchCore score meets the same objective (Section 4.4). Reporting both prevents surrogate-level compliance from being mistaken for evidence about the underlying model. When candidate production does not return enough query-satisfying alternatives, the system retains near-compliant candidates but marks unmet constraints explicitly and ranks them below fully compliant alternatives.

Candidate production compares two generators with one retrieval method. DiCE [11] produces diverse alternatives, while Actionable Counterfactual Explanations (ACFX) [47] uses explainable boosting machines to seek alternatives close to the factual configuration. LogTopK filters observed HPO trials by the query constraints and re-ranks them using the surrogate. It therefore cannot recommend configurations outside the log. The archived notebooks use the legacy identifier `CFNOW` for this procedure, but it is not the CFNOW tabu-search algorithm [14]. The comparison tests whether generation beyond the log adds value while holding surrogate scoring fixed. Each method receives the factual configuration, the surrogate model, the desired performance interval, the permitted feature ranges, and the required number of alternatives:

$$C_m = G_m(\mathbf{x}_0, \hat{f}_\theta, R, \mathcal{X}_q, K), \tag{13}$$

where G_m denotes the procedure used by method m.

Candidates are post-processed to remove invalid rows, enforce feature domains, and rank alternatives. Ranking prioritises objective satisfaction, soft-constraint satisfaction, predicted quality, and proximity to the factual configuration. Proximity is measured by a mixed-type distance:

$$d(\mathbf{x}', \mathbf{x}_0) = \sum_{j \in \mathcal{N}} \frac{|x'_j - x_{0j}|}{r_j} + \sum_{j \in \mathcal{C}} \mathbb{I}[x'_j \neq x_{0j}], \tag{14}$$

where $\mathcal{N}$ and $\mathcal{C}$ are the numerical and categorical feature sets, and r_j is the range of numerical feature j.

3.6. Conversational Query Interface

The conversational interface allows users to express configuration goals in natural language while keeping the recommendation process constrained and reproducible. A user request u is mapped to a structured query $q = \Phi(u)$, where Φ denotes the language-model-based parser followed by deterministic validation, repair, normalisation, and snapping. The parser is implemented with an instruction-tuned open-weight language model (Qwen2.5-3B-Instruct with Phi-3-mini and Llama-3.2-3B supported as alternatives).

The structured query has three components. The *objective* is `target_min`(v), `delta_improve`(δ), or `target_band`(L, U). The *soft constraints* give training fraction and review budget as `free`, `fixed`, or `range`, and corruption level as `free`, `fixed`, or `allowed`. The *selection* request is an integer K.

To illustrate the interface, a user asking *"Find me three configurations with F1 at least 0.85 and review budget below 0.30, training on no more than 60% of the data"* is mapped to `target_min(0.85)` with `review_budget: range(0, 0.30)` and `training_fraction: range(0, 0.60)` as soft constraints; a request for *"10% less review effort at similar quality"* is *representable* as `delta_improve(0.0)` with the relative constraint `review_budget: range(0, 0.90 * current_budget)`. We stress the distinction between what the schema can express and what the evaluated parser resolves: relative requests of this form are not reliably produced by the parser (Section 4.11) and currently trigger the explicit fallback. Deterministic resolution requires extracting the target field, direction, and percentage, then computing the bound in code; this is not yet implemented.

The language model is instructed to return only JavaScript Object Notation (JSON) matching the expected schema. The raw output is normalised to handle common shorthand forms, validated against the schema, repaired if necessary, and snapped to valid domains. For example, review-budget values are snapped to the allowed grid, training fractions are clipped and quantised to the valid range, and corruption levels are restricted to `none`, `mild`, or `strong`. Explicit numerical preferences detected in the user request are overlaid after parsing, so that user-stated values take precedence over ambiguous model interpretations.

The factual reference for generation is resolved automatically from the optimisation log: after scoring all trials with the surrogate, the interface selects n anchor trials (n=3 by default) combining the best-predicted trial, a median-budget trial from the upper predicted range, and greedily diversity-maximising additions. The validated query is applied at each anchor; per-anchor candidates are pooled, deduplicated, ranked by predicted quality, and truncated to the requested K.

The interface presents alternatives in a compact table (predicted quality, main configuration choices, soft-control values, constraint status), followed by a textual explanation intended to summarise the objective, compare alternatives, identify best and runner-up, report ties correctly, and surface shared violations once. These are design intentions of the explanation layer; we did not audit the factual faithfulness of the generated prose against the structured payload (Section 6.3). This confines the language model to translating intent into a structured query and verbalising a structured payload, while ranking, constraint checks, and candidates remain grounded in the surrogate, the feature domains, and the candidate-production method.

4. Technical Evaluation

This section evaluates the proposed counterfactual-guided HPO pipeline in the visual anomaly-detection setting. The evaluation has three goals: assessing whether the surrogate is accurate enough for fast local search, whether the methods return alternatives satisfying the structured operator-facing query, and whether selected recommendations survive confirmation by the underlying PatchCore pipeline. The pipeline is evaluated on MVTec, the controlled benchmark, and Proditec, the production use case, with the full three-method comparison and 24 confirmed candidates each, followed by the five diagnostic analyses announced in Section 1 (Sections 4.7–4.11).

4.1. Datasets and Use Cases

MVTec AD. In our evaluation, MVTec [2] provides controlled train/validation/test splits and a well-understood performance landscape, serving as the benchmark half of the two-dataset comparison. Our MVTec configuration log contains 9,976 completed trials over the PatchCore configuration space across four of the fifteen MVTec categories (bottle, cable, capsule, hazelnut). This restriction is a compute-budget decision fixed in advance rather than chosen on downstream results: every logged trial retrains PatchCore, so four categories alone cost roughly 85 hours of graphics processing unit (GPU) time (Section 4.10) and all fifteen would scale that by about 4×.

Proditec. Unlike MVTec, Proditec data are acquired under varying production conditions and reflect image variability, sensor noise, and imbalanced defect frequencies. This lets us test whether the pipeline generalises from benchmark to deployment and whether MVTec's confirmation behaviour holds in a different data regime. Our Proditec configuration log contains 2,999 completed trials over the same configuration space.

Both use cases share the same PatchCore configuration space (Table 2), enabling direct comparison of surrogate fidelity and main-model confirmation across datasets.

4.2. Experimental Protocol

All experiments ran on a single workstation with one NVIDIA RTX A5500 GPU (power-capped to 70% of its board limit), an AMD Threadripper PRO 5995WX central processing unit (CPU), and 256 gigabytes of memory. The experimental pipeline starts from Optuna optimisation logs collected independently over the PatchCore configuration space for each use case. Each completed trial stores the sampled hard hyperparameters, the soft controls, and the resulting validation F1 score. Invalid or incomplete trials are removed, categorical variables are normalised, numerical variables are snapped to their allowed domains, and the remaining trials are used to train the surrogate model $\hat{f}_\theta$ (Eq. 8) by minimising the squared prediction error of Eq. 9. The original stratified 80/20 split is retained as a development comparison: CatBoost uses its 20% portion for early stopping, so it is not treated as independent test evidence. The primary fidelity estimate uses the grouped cross-validation protocol of Section 4.3.

A CatBoost regressor [48] is used as the surrogate because it provides native handling of categorical configuration features and competitive predictive performance on the configuration–performance dataset. On each dataset, the surrogate is queried by three candidate-production methods: DiCE, LogTopK, and ACFX, each instantiating the call of Eq. 13. For each method we request $K = 8$ candidates, yielding 24 per dataset. This provides several alternatives while keeping full confirmation feasible. When query-satisfying candidates are insufficient, the fallback of Section 3.5 retains explicitly marked near-compliant

candidates. Candidates are filtered and ranked according to objective satisfaction, soft-constraint satisfaction, predicted quality, and distance from the factual configuration. A candidate satisfies its surrogate query if it meets Eq. 12 within the feasible region of Eq. 11. Since the technical evaluation has no live operator, the factual configuration $\mathbf{x}_0$ is a logged trial chosen by the experimenter from a pipeline-presented set containing the trials nearest the 50th, 70th, 85th, and 95th validation-F1 percentiles plus the best trial. The selected factual is held fixed across all three methods. For exact reproduction, `mvtec_all_methods_cfs_metadata.json` and `proditec_all_methods_cfs_metadata.json` record the primary `factual_index`, which resolves the complete feature vector in the released trial table; all three methods load that same payload. Both datasets use the same methods, K, and free soft controls. The recorded MVTec objective is `target_min(0.70)`; Proditec uses `target_band(0.75,0.95)` (Eq. 10).

Finally, every candidate is confirmed using the underlying PatchCore pipeline, which retrains the model under the candidate configuration on the corresponding splits and recomputes the actual validation and test F1 scores together with test area under the receiver operating characteristic curve (AUROC). All 24 candidates are confirmed on each dataset (48 retrainings), separating fast recommendation from main-model evidence. For the diagnostic analyses of Section 4.7, each confirmed candidate is re-run once with the same nominal seeds to capture the raw per-image anomaly scores. This is a linked diagnostic replication of the confirmation protocol, not a seed-variability test; the largest MVTec F1 difference between the two runs is 0.019.

Implementation details. PatchCore is our own implementation over torchvision feature extractors (`wide_resnet50_2`, `resnet18/34/50`, with ImageNet weights when `pretrained` is true) rather than a packaged library. Corruption is applied deterministically to training (normal) images only, seeded per image path: `mild` is a Gaussian blur of radius 0.6; `strong` is a Gaussian blur of radius 1.2 followed by additive Gaussian noise ($\sigma = 8$ on the 0–255 scale, clipped); `none` leaves images unchanged. Held-out images are divided into validation and test halves by a label-stratified 50/50 split. Proditec has 4,538 normal training images, validation 567 normal / 300 anomalous and test 568 / 299, with anomaly prevalence $\approx$ 34.5% in both; with the same construction applied per MVTec category. HPO uses Optuna's TPE sampler (fixed seed, `multivariate=False`, `group=False`), with 1,500 start-up trials for Proditec and 4,000 for MVTec, maximising Eq. 4, with no pruning and a fixed trial budget. The surrogate is CatBoost (5,000 iterations with early stopping after 200 rounds, learning rate 0.05, depth 8, `l2_leaf_reg` 3.0, root mean squared error loss). DiCE uses `method="random"` with per-feature ranges taken from the observed log domain; ACFX fits its required main-effects explainable boosting machine (EBM) classifier (`interpret`, `interactions=0`) to CatBoost-derived objective labels, and all final candidates are rescored with the common CatBoost surrogate, so the three methods share final scoring. Software: Python 3.11.9, PyTorch 2.10.0 (CUDA 12.8), torchvision 0.25.0, scikit-learn 1.6.1, Optuna 4.2.0, CatBoost 1.2.8, dice-ml 0.12, acfx 0.3.6, interpret 0.7.3, transformers 5.1.0.

Table 3: Development-split surrogate fidelity for MVTec and Proditec (CatBoost compared with HistGradientBoosting [49]). RMSE denotes root mean squared error; MAE, mean absolute error; and R^2, the coefficient of determination. Arrows indicate whether lower (↓) or higher (↑) is better. CatBoost uses this split for early stopping; grouped cross-validation below is the primary generalisation estimate.

Dataset	Model	RMSE ↓	MAE ↓	R^2 ↑
MVTec	CatBoost	0.015	0.008	0.996
MVTec	HistGradientBoosting	0.015	0.008	0.996
Proditec	CatBoost	0.086	0.046	0.936
Proditec	HistGradientBoosting	0.085	0.043	0.937

4.3. Surrogate Fidelity

We benchmarked 13 candidate regressors for the surrogate role (twelve scikit-learn models plus CatBoost); Table 3 reports the two most relevant: CatBoost, the operational surrogate, and HistGradientBoosting, the strongest alternative on both datasets. We selected CatBoost not because it topped the benchmark but because it handles the mixed categorical–numerical space natively for the downstream counterfactual pipeline and ships built-in feature-importance and Shapley tooling, so the surrogate's behaviour is directly inspectable as required by the paper's interpretability focus.

The surrogate thus captures the main structure of the configuration–performance relationship on both datasets. MVTec's higher R^2 partly reflects its narrower validation-F1 range (roughly 0.02–0.82 versus 0.0–0.91 on Proditec, where real production variability widens the target).

Because TPE revisits promising regions, the logs contain exact duplicate configurations (15.0% of MVTec rows, 17.6% of Proditec rows), so a shuffled split places some test configurations whose exact signature also appears in training (19.9% on MVTec, 26.2% on Proditec). We therefore use 5-fold GroupKFold cross-validation grouped by exact configuration signature as the primary generalisation check. It removes cross-fold duplicates and trains fixed 500-iteration models without holdout-driven early stopping. The resulting estimates (R^2 = 0.995 ± 0.0005 on MVTec, 0.934 ± 0.008 on Proditec) closely match the development values in Table 3, so the fidelity conclusion is not an artefact of duplicate leakage or reuse of the early-stopping split.

4.4. Surrogate Query Satisfaction and Main-Model Confirmation

Surrogate-level query satisfaction is 24/24 on both datasets: every candidate meets its recorded objective and soft constraints under Eq. 12. Main-model confirmation gives the complementary result. After retraining, 21/24 Proditec candidates meet the requested [0.75, 0.95] band, whereas 0/24 MVTec candidates meet the 0.70 minimum. The first figure verifies the structured search and constraint layer; the second measures whether the surrogate estimate persists in PatchCore. Section 4.7 analyses why these layers agree on Proditec but diverge for the primary MVTec query.

The methods differed substantially in proximity (the mixed-type distance of Eq. 14) and generation time, consistently across datasets (Table 4): DiCE was fastest with moderate distance, LogTopK fast but most distant, and ACFX two orders of magnitude slower but closest to the factual configuration.

4.5. Comparison of Candidate-Production Methods

Under the primary factual configuration, Table 4 summarises the generation and confirmation results for both datasets. Three observations stand out. First, the production evaluation confirms useful guidance: LogTopK is the best-performing Proditec method (confirmed F1 0.887, gap −0.011), DiCE is close behind (confirmed F1 0.806, gap −0.012), and even ACFX, the weakest on Proditec, reaches confirmed F1 0.751 with a gap of −0.051. Second, MVTec reveals a selection-conditioned reliability problem: LogTopK receives the highest surrogate scores (mean 0.815) but the lowest confirmed F1 (mean 0.569) and the largest gap (−0.246), while DiCE gives the best confirmed F1 (mean 0.623). Third, the proximity and runtime profiles are stable across datasets: DiCE is fastest with moderate distance, LogTopK is fast but distant, and ACFX is slowest but closest to the factual. Because the two datasets use different objectives, their ranking reversal is descriptive rather than a controlled dataset effect.

Confirmed PatchCore F1 is stable within methods, as the standard deviations in Table 4 show; LogTopK is the tightest on both datasets (0.003 and 0.005 over eight retrainings each). Corresponding gap standard deviations lie between 0.003 and 0.024, and complete per-candidate variability tables are included in the public archive.

4.6. Main-Model Confirmation

The central technical result is the contrast between successful production confirmation and a benchmark reliability boundary. On Proditec, the 24 recommendations average confirmed PatchCore test F1 of 0.815, a mean surrogate-to-model gap of only −0.024, and test AUROC of 0.947; 21 satisfy the requested target band. The same evaluation protocol exposes different behaviour on MVTec: mean confirmed F1 is 0.599, the gap is −0.190, and pooled test AUROC is 0.497. Proditec therefore demonstrates that the framework can deliver useful recommendations in the studied production setting, while MVTec shows why aggregate surrogate performance cannot by itself authorise an individual recommendation. Main-model confirmation remains the final deployment gate in both settings.

Figure 2 plots confirmed PatchCore F1 against surrogate F1 for every candidate on Proditec: most points lie near the ideal-agreement line, with the tightest cluster for LogTopK in the high-quality region and the largest individual gaps among ACFX candidates.

A complementary view on Proditec compares the soft-parameter profiles of the six candidates with the highest confirmed PatchCore F1 against the remaining 18. The top confirmed candidates obtain a mean confirmed PatchCore F1 of

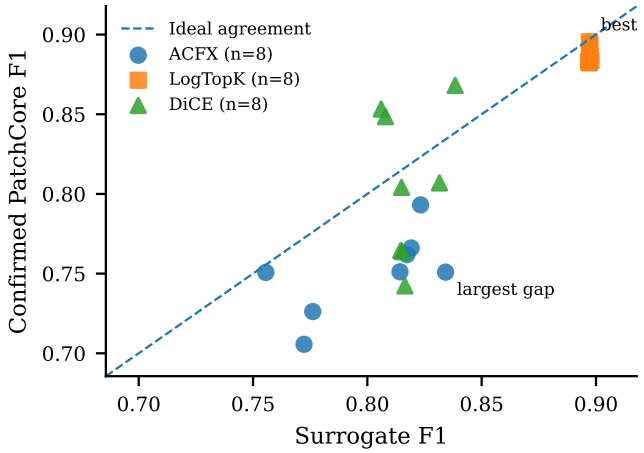


Figure 2: Confirmed PatchCore F1 versus surrogate F1 for all 24 Proditec candidates (best candidate and largest gap annotated). The archived chart label CFNOW denotes LogTopK.

0.888 versus 0.790 for the others, using a *lower* average training fraction (0.792 compared with 0.832), a lower review budget (0.345 compared with 0.371), and, most strikingly, uniformly `none` or `mild` corruption, whereas the weaker group is dominated by `strong` corruption. A standardised regression of confirmed F1 on the three soft controls jointly over the 24 candidates supports this reading beyond the post-hoc top-6 split: the strong-corruption indicator is the only significant term (coefficient −0.048, $p \approx 10^{-4}$; Spearman $\rho = -0.77$; model $R^2 = 0.67$), while training fraction and review budget are not significant. The corruption assumption is thus the soft control most strongly associated with confirmed quality on Proditec, and the best configurations are not those that consume the most training data. This is exactly the kind of trade-off that a standard HPO ranking would hide and that the operator-centred view is designed to surface.

Section 4.7 decomposes this dataset difference at the level of raw anomaly scores, separating surrogate error on the defined target from threshold transfer and from cross-category score-scale effects.

4.7. Diagnosing the Confirmation Gap

To move beyond aggregate gaps we re-ran each of the 24 confirmed candidates per dataset with the same nominal seeds, capturing raw per-image anomaly scores on validation and test, and analysed thresholding, category, and distance effects without further retraining (Table 5).

Within-category ranking is intact on MVTec. Although the mean pooled test AUROC of the retrained MVTec candidates is at chance level (0.497), the mean *per-category* test AUROC of the very same models is 0.827: within each of the four categories, the retrained candidates rank anomalies well (Figure 3). Two distinct effects follow from pooling. First, because the anomaly-score scales differ across categories, merging them destroys the *global ranking*: pooled AUROC falls to chance even though no threshold is involved, a purely ranking-level

Table 4: Candidate-production and confirmation results (eight candidates per method; PatchCore F1 as mean ± standard deviation). Surrogate-level query satisfaction was 100% in all six cells and is therefore stated here rather than tabulated. Gap = PatchCore F1 minus surrogate F1. Distance is the mixed-type measure of Eq. 14 summed over the 13 configuration features, so larger values are further from the factual. The two datasets used different objectives; figures are not comparable across the horizontal rule. The archived code labels LogTopK as `CFNOW` (Section 3.5).

Dataset	Method	Objective	Distance	Time (s)	Surrogate F1	PatchCore F1	Gap
MVTec	ACFX	minimum 0.70	0.91	99.84	0.779	0.607 ± 0.023	−0.172
MVTec	LogTopK	minimum 0.70	5.65	5.14	0.815	0.569 ± 0.003	−0.246
MVTec	DiCE	minimum 0.70	1.58	0.36	0.773	0.623 ± 0.017	−0.150
Proditec	ACFX	band 0.75–0.95	0.80	111.37	0.802	0.751 ± 0.026	−0.051
Proditec	LogTopK	band 0.75–0.95	3.82	1.49	0.897	0.887 ± 0.005	−0.011
Proditec	DiCE	band 0.75–0.95	1.55	0.34	0.818	0.806 ± 0.047	−0.012

Table 5: Score-level diagnosis of the confirmation gap (means over the 24 confirmed candidates per dataset). "Deployed threshold" is the validation-calibrated pooled review-budget threshold used in the confirmation protocol; the oracle is the best single global threshold on the test scores; category-wise thresholds apply the same review budget within each MVTec category. The oracle row is test-fitted headroom, not the validation-to-test transfer reported in Section 4.7 (+0.013 on MVTec).

Quantity (mean over 24 candidates)	MVTec	Proditec
Surrogate-predicted F1	0.789	0.839
Confirmed F1 at deployed threshold	0.599	0.815
Confirmed F1 at oracle global threshold	0.832	0.839
Oracle headroom (oracle − deployed)	0.233	0.024
Pooled test AUROC	0.497	0.947
Per-category test AUROC	0.827	–
F1 gain from category-wise thresholds	+0.248	–

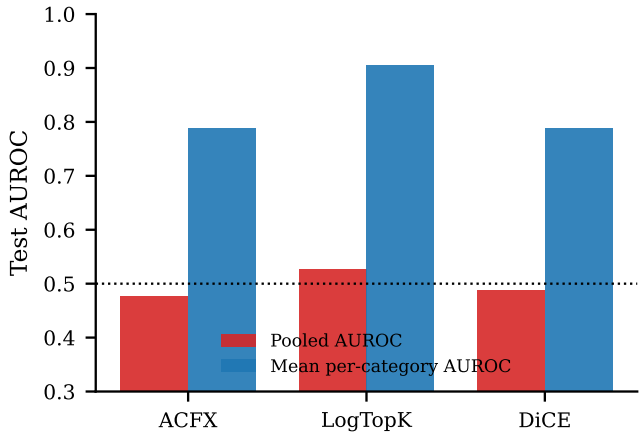


Figure 3: Pooled versus mean per-category test AUROC of the confirmed MVTec candidates by method. The retrained models rank anomalies well within each category; pooling categories whose anomaly-score scales are not commensurable destroys the global ranking, driving pooled AUROC to chance independently of any threshold.

effect (0.827 → 0.497, a penalty of 0.330). Second, and separately, a single validation-calibrated budget threshold applied to those incompatible scales over- or under-flags whole categories, which damages *F1*: applying the same review budget *within* each category recovers +0.248 mean F1 without changing the models. The ranking and threshold effects must therefore be interpreted separately because AUROC is threshold-independent.

The discrepancy appears before validation-to-test transfer. The surrogate's prediction target is pooled validation F1 under the review-budget protocol (Eq. 4), so the gap must first be split at the validation boundary. For the MVTec candidates the retrained pooled validation F1 is 0.585±0.023 against a mean surrogate prediction of 0.789 ± 0.023: a surrogate error of −0.204 on the very quantity the surrogate is trained to predict. The subsequent validation-to-test transfer at the deployed threshold is small (+0.013 ± 0.010). The total confirmation gap of −0.190 is therefore dominated by the prediction-to-rerun-validation term on the recommended configurations, not by validation-to-test threshold transfer; on Proditec both components are small (surrogate term −0.009 ± 0.031, transfer −0.015 ± 0.020). This revises the natural first reading of Table 4: a surrogate with development R^2 = 0.996 and grouped R^2 = 0.995 can still be badly wrong on the specific configurations a constrained query selects.

Pooled score-scale incompatibility is a deployment bottleneck. The surrogate learns one pooled-validation F1 value per configuration, but that scalar target does not preserve the category-specific score distributions from which the pooled metric is produced. In the upper region of the MVTec log, configurations happen to pool coherently; query-targeted candidates generated around the selected mid-log factual do not. The result is configuration-dependent rather than universal: the six additional upper-quantile MVTec candidates of Section 4.9 retrain to pooled test AUROC ≈ 0.98 with gaps within ±0.02. Moreover, sweeping the review budget over its full grid does not repair the affected candidates (best budget improves confirmed F1 by at most 0.017), so no choice of β compensates for incommensurable scales. Together with the per-category analysis, these observations implicate the pooled objective and calibration protocol as the operational bottleneck. They do not establish score-scale mismatch as the sole cause of every surrogate prediction error.

An oracle threshold bounds discrimination, and nothing more. An oracle global threshold fitted on the *test* labels raises mean confirmed MVTec F1 from 0.599 to 0.832 (Proditec: 0.839). This is an upper bound on what a test-calibrated rule could extract, not an achievable deployment result, and it does not indicate that the surrogate predicted its target correctly.

Distance and configuration degeneracy are secondary. Across all 24 MVTec candidates, the gap correlates strongly with the mixed-type distance of Eq. 14 (Pearson $r = -0.91$, $p < 0.001$), but this association is driven by differences *between* methods (the distant LogTopK candidates carry the largest gaps); within each method the correlation is weak and not significant (ACFX $r = -0.46$, LogTopK $r = -0.02$, DiCE $r = -0.54$; all $p > 0.05$). On Proditec the pooled correlation is even positive ($r = 0.58$, $p = 0.003$): larger distances do not produce larger gaps. The worst-gap candidates show no configuration degeneracy on MVTec (all 24 use pretrained weights, none strong corruption or an extreme coreset ratio), while Proditec's modest residual gaps concentrate in `resnet34`/single-layer candidates under `strong` corruption (mean gap −0.035 over 14 candidates versus −0.011; AUROC 0.931 versus 0.970).

Consequence: evaluate fidelity where decisions are made. Aggregate cross-validation measures fidelity over the logged configuration distribution, whereas a recommendation system acts on a subset selected by its surrogate and the operator's constraints. We refer to accuracy on that induced subset as *selection-conditioned fidelity*. The MVTec result shows why it must complement global R^2: reliability is poor for the primary query-selected region but strong for the additional upper-quantile region. Improving aggregate surrogate fit alone would not address the separate deployment problem created by pooling incompatible category scales. The supported design rules are therefore to test fidelity on configurations selected by representative queries, confirm final candidates with the underlying model, and define and calibrate review-budget objectives per category for heterogeneous product types.

4.8. Sensitivity to the Factual Configuration

The main method comparison (Section 4.5) uses one factual configuration per dataset. As an auxiliary anchor-sensitivity analysis, we repeated generation under a common target-band query (`target_band` $[0.75, 0.95]$, free soft controls, $K = 8$) at five factual anchors drawn from each log to span the target distribution (the trials nearest the 50th, 70th, 85th, and 95th percentiles of validation F1, plus the best trial), covering the two fast methods, DiCE and LogTopK (80 additional surrogate-level candidates per dataset, no additional retraining); ACFX was excluded because of its two-orders-of-magnitude higher generation cost.

Three properties are stable across all five anchors on both datasets. First, surrogate-level query satisfaction remains 100% in every anchor–method combination, so the post-processing and validation layer is robust to the starting point. Second, under this auxiliary query, LogTopK's mean surrogate F1 exceeds DiCE's at every anchor, showing stable ordering across the tested factual configurations. Third, the methods behave as their designs suggest: LogTopK returns the same top-scoring logged configurations at every anchor (constant $\bar{F}_1$ 0.815 on MVTec, 0.897 on Proditec), so its distance to the factual varies purely with the anchor's position (1.41–5.65 and 0.18–3.83), whereas DiCE adapts locally, trading a lower score near weak anchors ($\bar{F}_1$ 0.781–0.814 and 0.832–0.893) for smaller distances near strong ones (down to 0.69 and 0.09). This sweep is surrogate-level only; confirmation across anchors remains future work (Section 6.3).

Table 6: Confirmed generation-versus-retrieval comparison under the tight quantile band (top two candidates per method; means of surrogate and confirmed PatchCore F1 and their gap).

Dataset	Method	Surrogate F1	Confirmed F1	Gap
MVTec	DiCE	0.802	0.793	−0.009
MVTec	LogTopK	0.808	0.793	−0.015
MVTec	NearestFeasible	0.800	0.801	+0.000
Proditec	DiCE	0.896	0.876	−0.020
Proditec	LogTopK	0.898	0.888	−0.010
Proditec	NearestFeasible	0.891	0.882	−0.009

4.9. Generation versus Log Retrieval under Constrained Queries

We tested generation beyond the log with twelve structured queries (ten base and two dataset-relative quantile variants) at three factual anchors per dataset. They cover broad, tight, absolute, quantile, one-sided, relative, combined, and infeasible constraints. We compare DiCE with LogTopK, which ranks query-satisfying logged configurations by surrogate score, and NearestFeasible, which ranks them by the mixed-type distance of Eq. 14. For DiCE we report raw counts before repair or domain snapping and final query satisfaction. Because LogTopK is surrogate-ranked, this experiment tests generation beyond the log, not surrogate ranking against the recorded trial F1.

Two findings summarise the suite. First, in every query cell where any method produced query-satisfying candidates, the retrieval baselines did as well, and DiCE satisfied no query that retrieval could not; conversely, the absolute tight band $[0.85, 0.90]$, the relative budget-reduction query, and the combined tight query were infeasible on MVTec for *all* methods, generation included, because the surrogate assigns no configuration a score in the requested region. Second, main-model confirmation of the top candidates under the tight quantile band (two per method per dataset, twelve additional PatchCore retrainings) shows retrieval matching or slightly exceeding generation in confirmed quality (Table 6). All six MVTec candidates, drawn from the upper-quantile region of the log, retrain to pooled test AUROC ≈ 0.98 with gaps within ± 0.02 (compare Section 4.7).

Across this suite, off-log generation does not increase feasible-query coverage, and confirmation under the tight quantile band gives comparable top-candidate quality. We read this as a coverage-conditioned deployment result rather than a general ranking of counterfactual methods: for the two logs studied here, retrieval is the appropriate first-stage policy because it stays within previously evaluated configurations without sacrificing coverage or confirmed quality. Generation remains useful when retrieval returns too few feasible, local, or diverse alternatives, a regime not observed in these dense logs that should be tested under deliberately reduced coverage. Because logged status did not guarantee confirmation under every query, a retrieve-first policy reduces unsupported search but does not replace main-model confirmation.

Table 7: Component and complete-interaction wall-clock costs. The eight-retraining and interaction-plus-confirmation rows are derived from the measured means to provide matched multi-alternative comparators.

Quantity	MVTec	Proditec
PatchCore retraining (s)	30.8	67.1
Surrogate prediction (ms)	2.15	2.05
Component speed-up	14,306×	32,720×
Sequential eight retrainings (s)	246.4	536.8
Eight-candidate interaction (s)	141.4–241.2	141.4–252.8
Interaction + one confirmation (s)	172.2–272.0	208.5–319.9

4.10. Computational Cost and Reuse

We distinguish the cost of one surrogate prediction from the complete interaction. Table 7 compares mean PatchCore confirmation time, surrogate latency over 200 warmed-up single-row queries, and the measured CPU-only interaction including parsing, candidate production, and explanation. The complete-interaction range sums measured parse, anchor-scoring, generation, ranking, and explanation times per method; the two language-model stages were timed over the 40 labelled utterances of Section 4.11. Their dataset-independent cost sets the common lower bound, while method-specific generation sets the upper bound. No power telemetry was collected, so energy and CO_2 estimates are not reported.

The isolated surrogate component is four orders of magnitude faster than one retraining, but parsing, candidate production, and explanation dominate complete-interaction latency. One eight-candidate interaction therefore takes longer than one PatchCore retraining. For the intended screening task, however, the matched unit is a set of alternatives. At the measured mean costs on the evaluated single-workstation hardware, sequentially retraining eight configurations would take 246.4 s on MVTec and 536.8 s on Proditec, whereas the interaction screens eight in 141.4–241.2 s and 141.4–252.8 s, respectively. This corresponds to a conditional screening advantage of 1.02–1.74× on MVTec and 2.12–3.80× on Proditec. If one selected candidate is then confirmed, the total remains below eight sequential retrainings for all Proditec routes and for the faster MVTec candidate producers, but not for the slowest MVTec route. This is an amortised wall-clock result for multi-alternative screening, not an equivalence between surrogate predictions and confirmed scores. No energy or CO_2 conclusion is drawn without power telemetry.

4.11. Evaluation of the Conversational Parser

The utterance-to-structured-query pipeline was previously assessed only end-to-end through the user study. We add a direct evaluation on 40 labelled operator utterances covering all three objective types, absolute and relative constraints, percentage and decimal units, alternative-count extraction (including number words), fixed-value snapping, decoy numbers, paraphrase pairs, and deliberately infeasible or out-of-domain requests. Two front-ends are scored against field-level gold annotations: the one deployed during the user study, and the released version, which adds a factual-aware prompt with worked examples, coercion of one-sided constraint phrasings, and one error-aware retry, keeping schema, validator, repair, and fallback layers identical. Scoring is exact at the field level after canonical snapping, with a documented semantic-equivalence variant that credits operationally identical encodings; fallback outputs receive no credit under either scoring.

Table 8: Labelled parser evaluation (40 utterances). "Accepted@1" = model JSON schema-valid on the first attempt; fallback outputs score zero. The user study ran on the study-time front-end; the revised front-end is the released version.

Metric	Study-time	Revised
Accepted@1 (%)	12.5	72.5
Fallback rate (%)	87.5	22.5
Objective type accuracy (%)	35.0	77.5
Objective value accuracy (%)	12.5	70.0
Constraint fidelity, train fraction (%)	7.5	75.0
Constraint fidelity, review budget (%)	10.0	72.5
Constraint fidelity, corruption (%)	7.5	75.0
K extraction accuracy (%)	82.5	87.5
Exact full query (%)	5.0	60.0
Exact | accepted (%)	40.0	77.4
Exact, relative subset (%)	0.0	0.0
Exact, paraphrase subset (%)	16.7	66.7

Table 8 summarises the results. The study-time front-end produced schema-valid JSON on the first attempt for 12.5% of utterances, fell back for 87.5%, and achieved 5% exact full-query accuracy. The revised front-end raises first-attempt acceptance to 72.5%, cuts fallback to 22.5%, and reaches 60% exact (62.5% semantic) accuracy; exact accuracy among accepted JSON outputs is 77%. Greedy decoding makes repeated parses identical. Both front-ends score 0% on relative constraints such as "10% less review effort": the three-billion-parameter model cannot reliably perform arithmetic against the current configuration, so these requests trigger the explicit fallback. This matches the user study, where condition-B participants needed most attempts on baseline-relative Task 3 (6.3, compared with 1.6 and 3.5 on absolute Tasks 1 and 2) because iteration, anchoring, and overlay resolved relative requests. Although the released front-end handles absolute requests substantially better, 40% of labelled utterances are still not parsed exactly. The parser should therefore extract the relative field, direction, and quantity, then resolve the arithmetic deterministically before candidate production.

5. Evaluation with Human Participants

The proposed framework is intended to support human decision-making during configuration selection, not only to generate query-satisfying counterfactual alternatives. We therefore conducted a user evaluation to examine whether participants could select machine-learning configurations under quality and deployment-oriented constraints, and whether the conversational interface supported comparison of alternatives and understanding of configuration trade-offs. The study was run with computer-science master's students rather than industrial operators; it therefore tests whether the interface makes constraint-based configuration selection tractable for technically literate non-specialists, which is a necessary but not suffi-

cient condition for deployment with production operators (Section 6.3).

5.1. Study Design

The study used a between-subjects design with two conditions: (A) an application programming interface (API) baseline, using the pipeline through its Python API in a Jupyter notebook with criteria in code, and (B) the conversational interface, with natural-language criteria and textual explanations alongside tabular results. Both conditions used the same pipeline: anchors were selected from the log (Section 3.6), candidates were generated under the constraints, scored by the surrogate, and merged into a presented top-K. The conditions differed only in how criteria were expressed and results presented.

Under the university requirements applicable to this minimal-risk, course-based software-usability evaluation, no separate ethics-committee review was required. The responsible academic supervisor and the course coordinator authorised the study. Participation was voluntary and unpaid, did not affect course assessment, and participants could withdraw at any time. All participants received study information and provided written informed consent. Each session began by explaining that the tool, not the participant, was evaluated. Participants then completed three configuration-selection tasks. In each task, they selected one final configuration and wrote a short justification explaining their choice. After completing the tasks, participants answered a short post-task questionnaire about their experience with the assigned system.

Tasks 1 and 2 evaluated selection under *absolute* constraints on predicted F1 score, training fraction, review budget, and corruption level. Task 3 evaluated selection under constraints defined *relative* to a base configuration: participants had to select an alternative that preserved predicted F1 score while reducing review budget by at least 5%. Table 9 summarises the task requirements.

The evaluation was exploratory, covering valid selection, completion time, perceived ease, and the Task-3 quality–effort trade-off, with descriptive statistics only, given the sample size.

5.2. Participants and Measures

Twenty-two computer-science master's students participated (one condition-A participant's duplicated session was removed from the logs, leaving 22 unique participants across 10 API and 12 conversational-interface sessions). All were enrolled in a computer-science master's programme and had completed coursework in machine learning, so they were familiar with model training and hyperparameters but were not industrial inspection operators; Section 6.3 discusses what this proxy population does and does not license. Participants were recruited during laboratory sessions; none was an author or involved in developing the system. Participants were randomly assigned to conditions. Beyond programme enrolment and prior machine-learning coursework, no demographic data were collected, which we note as a limitation (Section 6.3).

A task answer was counted as *usable* if the session log contained an explicitly marked final configuration selection for that task. The cleaned task-level dataset contained 65 usable final answers: 30 from condition A and 35 from condition B. Condition A contained 10 usable answers for each task; condition B contained 12 for Tasks 1 and 2 and 11 for Task 3, because one participant completed the session without recording a final Task 3 selection.

After removing one response without a valid identifier and one duplicate (later submission kept), 20 valid questionnaire responses remained (9 condition A, 11 condition B); the shortfall from 22 reflects voluntary completion. The questionnaire contained five Likert-scale items (1–5, higher is more positive): ease of identifying valid F1 configurations, checking soft constraints, understanding trade-offs, comparing options, and using the provided tool. Together with the overall small sample size, the mild imbalance (9 compared with 11) means that Likert comparisons should be treated as descriptive indicators rather than reliable effect estimates.

For each final answer, we recorded requirement satisfaction, the number and type of violated constraints, the number of attempts, and the written justification; for Task 3, additionally the base configuration, selected alternative, quality change, and review-budget reduction. Attempt counts are descriptive because interaction steps were logged differently across environments.

5.3. Results

Completion time. Completion times were similar (A: 23.64 min, standard deviation 14.69; B: 23.31 min, standard deviation 8.90); condition B's lower variability suggests a more consistent interaction path, though the sample is small.

Task validity. Table 10 summarises task-level validity. Both conditions produced high validity overall: condition A reached 90.0% on all three tasks, condition B 100.0% on Task 2 but less on Tasks 1 and 3. Task 3 was hardest in both conditions, requiring simultaneous quality preservation and a relative budget reduction. Because the all-answer means include invalid selections, they can fall outside a task's own bounds (condition A, Task 1: 0.773 with a 0.80 lower bound); the valid-only column gives the comparable solution-quality figures, under which the two conditions are within 0.02 of each other on every task.

Violation analysis. In condition B, Task 1 errors were caused by training fraction being outside the allowed range, suggesting that participants occasionally misread or deprioritised this constraint when focusing on F1 and review budget. In Task 3, the main issue was failure to satisfy the relative review-budget reduction requirement: two selections did not reduce the review budget by at least 5%, and one selection had lower predicted quality than the base configuration. In condition A, violations were distributed across quality, training fraction, review budget, and the same-or-better-quality requirement in Task 3.

Task 3 baseline-relative analysis. Task 3 was analysed separately because, unlike Tasks 1 and 2, its constraints are defined relative to a base configuration, the task form that most closely mirrors how counterfactual guidance would be used in practice.

Table 9: Configuration-selection tasks used in the user evaluation.

Task	Type	Requirement
Task 1	Independent constraints	Select one configuration with predicted F1 score between 0.80 and 0.92, training fraction between 0.40 and 0.70, review budget at most 0.40, and any corruption level allowed.
Task 2	Independent constraints	Select one configuration with predicted F1 score between 0.78 and 0.91, training fraction at most 0.60, review budget at most 0.30, and corruption level equal to `none` or `mild`.
Task 3	Baseline-relative constraints	Select one alternative compared with a base configuration. The selected alternative must have predicted F1 score not lower than the base configuration and review budget at least 5% lower than the base configuration.

Table 10: Task-level validity and selected-configuration summaries. "Mean F1 (all)" averages every recorded final answer including invalid ones; "Mean F1 (valid)" averages valid selections only.

Condition	Task	n	Valid (%)	Violations	Mean attempts	Mean F1 (all)	Mean F1 (valid)	Mean budget	Training share
API (A)	Task 1	10	90.00	0.200	1.600	0.773	0.851	0.340	0.581
API (A)	Task 2	10	90.00	0.200	1.200	0.831	0.822	0.264	0.379
API (A)	Task 3	10	90.00	0.100	1.200	0.875	0.879	0.318	0.880
Conversational (B)	Task 1	12	83.33	0.167	1.583	0.851	0.853	0.370	0.543
Conversational (B)	Task 2	12	100.00	0.000	3.500	0.818	0.818	0.271	0.349
Conversational (B)	Task 3	11	81.82	0.273	6.273	0.883	0.896	0.361	0.895

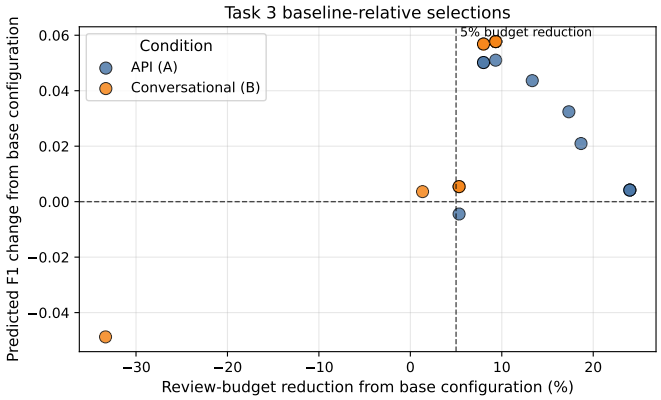


Figure 4: Task 3 quality–effort trade-off relative to the base configuration.

The base configuration had predicted F1 = 0.850 and review budget = 0.375. Relative to it, condition A achieved mean $\Delta F1 = 0.026$ with a 15.20% budget reduction (90.0% valid, $n = 10$) and condition B $\Delta F1 = 0.033$ with 3.76% (81.82% valid, $n = 11$): condition B gained slightly more predicted quality while condition A cut review effort far more.

Figure 4 visualises the quality–effort trade-off for Task 3. Valid selections should fall above $\Delta F1 = 0$ and to the right of the 5% budget-reduction threshold. The figure shows that several condition-B participants preserved or improved predicted quality but did not achieve sufficient budget reduction, suggesting that the interface's emphasis on quality maximisation came at the expense of satisfying the relative budget constraint.

Questionnaire. Condition B scored higher on four of the five post-task questionnaire items. The largest difference was for comparing options (B: 4.36 compared with A: 3.44), followed by identifying valid F1 configurations (4.91 compared with 4.44) and understanding trade-offs (3.91 compared with 3.56); the two conditions were close on checking constraints (4.18 compared with 4.00) and on using the provided tool, where condition A was marginally higher (4.33 compared with 4.27). Given the sample imbalance and small group sizes, these differences should be treated as indicative of perceived usability differences rather than as reliable effect estimates.

5.4. Qualitative Analysis of Participant Justifications

To complement the quantitative results we analysed the short written justifications qualitatively, using a large language model as an automated judge [50] to reduce confirmation bias. Anonymised justifications with condition, task, and selected-configuration metadata were submitted under a structured prompt covering attribute references, trade-off reasoning, condition differences, Task-3 what-if reasoning, and over-reliance on predicted F1, requiring descriptive analysis without overclaiming or identifying participants. The authors reviewed the summary for factual consistency with the raw comments; the model may still miss subtle patterns, and the output reflects the prompt's framing.

Overall pattern. Across conditions and tasks, participants grounded choices in a small set of observable attributes: predicted F1, review budget, and, less often, training fraction and corruption level. They provided brief, outcome-oriented justifications (most often highest F1 or lowest budget) and a minority engaging in deliberative trade-off reasoning; justification length was marginally higher in condition B.

Condition differences and design implications. The two conditions produced similar strategies with different framing: condition-A participants anchored on F1 maximisation, while condition-B participants more often referenced named options and the interface's "balanced compromise" framing, occasionally prioritising corruption level over F1. Three tentative design implications follow: secondary metrics (budget reduction, corruption robustness) should be scaffolded as first-class criteria; baseline-relative constraints need salient, continuously updated satisfaction feedback; and evaluative option labels should be used with care, since they may be accepted uncritically.

Task 3 and baseline-relative reasoning. Task 3 elicited the most varied justifications: the *relative* budget constraint led to at least two invalid selections believed to be valid, and several participants in both conditions optimised absolute F1 gain or the absolute budget value rather than the percentage reduction, suggesting that relative framing is cognitively less accessible than absolute comparisons.

6. Discussion

6.1. Interpreting the Two-Dataset Technical Results

The paired evaluation establishes a useful production result and the condition needed to trust it. On Proditec, 21/24 recommendations meet their target after retraining, with mean confirmed F1 of 0.815, gap of −0.024, and AUROC of 0.947. On MVTec, all 24 recommendations satisfy the surrogate query but the underlying model does not confirm the target. The −0.204 prediction-to-rerun-validation term and +0.013 validation-to-test term localise the discrepancy to the validation side rather than to test transfer. At the same time, per-category AUROC of 0.827, pooled AUROC of 0.497, and a +0.248 F1 gain from category-wise thresholds show that the pooled objective is unsuitable for the heterogeneous score scales in this deployment. Upper-quantile gaps within ±0.02 further show that reliability is selection-conditioned rather than uniformly poor. The different primary objectives prevent a controlled cross-dataset comparison, but together the experiments support a general evaluation rule: global surrogate fidelity must be complemented by confirmation on the region induced by representative operator queries.

6.2. Implications for Operator-Centred HPO

The user evaluation concerns the complete study-time workflow. Technically literate non-specialists produced high task validity in both conditions and reported greater ease of comparing options with the conversational interface. Because the study-time parser achieved only 5% exact full-query accuracy, these task results cannot be attributed to one-shot parsing; they reflect the combined interaction, including repair, anchoring, presentation, fallback, and iteration. The present study does not isolate the contribution of each layer, but it shows why parser accuracy and workflow task success must be evaluated separately. Baseline-relative requirements were hardest in both conditions, and quality maximisation sometimes displaced the requested budget reduction. The labelled parser test identifies a direct remedy: treat relative constraints as first-class objects, extract their field, direction, and quantity, and resolve the arithmetic deterministically. The revised parser and the target population of production operators are the next evaluation setting.

6.3. Limitations

Scope of datasets and task. The evaluation covers one benchmark (four MVTec categories) and one production use case on a single architecture (PatchCore) with a fixed configuration space. The score-scale pathology is observed in the pooled four-category MVTec setting and not in Proditec; two datasets do not establish that this difference generalises to all multi-category and single-product deployments.

Confirmation coverage. Confirmation covers all 24 main candidates per dataset plus the 12 retrieval-comparison candidates, but the multi-anchor sweep is surrogate-level only and per-candidate variability rests on single confirmations with one seeded re-run each.

Language-model parsing reliability. The released front-end parses 60% of labelled utterances exactly (77% among accepted JSON outputs) and falls back on 22.5%. Relative constraints fail entirely at the parser level. This is an addressable architecture limitation rather than a reason to delegate arithmetic to a larger language model: the parser can extract the quantity and reference field, after which deterministic code computes the bound. The user study used the earlier front-end (5% exact accuracy), so its results apply to the complete workflow, including repair, anchoring, overlay, fallback, and iteration, rather than to one-shot parsing.

User-study population. The participants were computer-science master's students, not industrial inspection operators. They are a proxy for reasoning about model quality, training data, and thresholds without being specialists in the deployed pipeline. Recruiting comparable numbers of production operators was not feasible for this study. They nonetheless differ from the target users in ways that plausibly matter: operators carry domain priors about defects and inspection cost, face real consequences for a wrong configuration, and may weigh review effort differently. The study therefore supports usability claims for technically literate non-specialists and motivates, rather than substitutes for, a planned operator-population study.

User-study scale and design. The user study is exploratory: 22 participants across two conditions, descriptive statistics only, condition-specific logging of attempt counts, and voluntary questionnaire completion with a mild response imbalance. Demographic attributes beyond programme enrolment and prior machine-learning coursework were not collected, so their influence cannot be analysed. The results should be read as design evidence, not as confirmatory effect estimates.

No surrogate support indicator. The MVTec results make the absence of a per-candidate reliability signal a substantive gap. The interface already computes mixed-type distance to the factual and logged status, so it could flag weakly supported candidates and recommend confirmation. Logged status alone is insufficient because the primary MVTec LogTopK candidates also show large gaps; a useful signal must combine support, local error, and target stability. Such a signal is not part of the evaluated system.

Explanation faithfulness unevaluated. The parser is evaluated, but the generated textual explanations are not. Because they are produced from a structured payload, a mechanically checkable faithfulness suite (agreement with the candidate table, correct constraint status, tie preservation, no unsupported trade-offs) is feasible and is planned; until then, the explanations should be regarded as unverified presentation rather than validated evidence.

Energy measurement. No component or end-to-end power telemetry was collected. Runtime is reported in Section 4.10, but no energy or CO_2 saving is claimed.

7. Conclusion

This paper presented CF-XHPO, a framework that turns HPO logs into operator-centred what-if guidance for industrial visual anomaly detection. It combines surrogate modelling, constrained retrieval, counterfactual generation, deterministic query validation, and a conversational interface. The evaluation confirms 60 selected configurations with PatchCore and demonstrates practical value on the Proditec production dataset: 21/24 primary recommendations meet their target after retraining, with mean confirmed F1 of 0.815, mean gap of −0.024, and test AUROC of 0.947.

The evaluation also establishes how such guidance should be validated. All 48 primary recommendations satisfy their surrogate-level queries, but the underlying model confirms 21/24 on Proditec and 0/24 for the primary MVTec query. MVTec therefore exposes selection-conditioned surrogate error that global R^2 does not reveal. The score-level analysis localises the discrepancy to the validation side and shows a separate deployment problem: incompatible category score scales reduce pooled AUROC from 0.827 per category to 0.497 and make one pooled review-budget threshold unsuitable. The resulting policy is concrete: evaluate fidelity on query-selected regions, calibrate heterogeneous categories separately, and confirm final recommendations with the underlying model.

The supporting analyses translate these results into system-design choices. Constrained retrieval matches off-log generation on the tested dense logs, motivating a retrieve-first policy with generation reserved for insufficient coverage, proximity, or diversity. Complete interaction overhead is substantial but amortises across an eight-candidate screen, producing a clear wall-clock benefit on Proditec and on the faster MVTec routes under a sequential-retraining comparator. In the user study, both workflows support high task validity, while conversational participants report greater ease of option comparison. The parser evaluation separates this workflow-level result from one-shot parsing and identifies deterministic resolution of relative constraints as the immediate interface improvement. Together, these findings position CF-XHPO not as an automatic replacement for model evaluation, but as an empirically validated screening and decision-support layer over trials that an HPO process has already produced.

Future work will define and calibrate the review-budget objective per category, resolve relative constraints deterministically before generation, surface a support indicator distinguishing logged from generated candidates, evaluate the factual faithfulness of the generated explanations, repeat the user evaluation with the released parser and with production operators, and extend confirmation across anchors, seeds, and further industrial use cases.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted Technologies

During preparation, the authors used Claude Opus 5 (Anthropic) to support the implementation and analysis of experiments; all analyses were verified by the authors against the released code and results before inclusion in the manuscript. Claude Opus 5 was also used to support the qualitative summarisation of anonymised participant comments (Section 5.4) and to improve language and readability. The authors reviewed and edited all output and take full responsibility for the publication.

Data availability

The MVTec AD benchmark is publicly available [2]. The Proditec dataset is proprietary industrial data and cannot be shared. All analysis code and derived data, together with a live demonstration of the conversational interface, are publicly archived; the repository and demonstration links are withheld here for double-blind review, are available to the editor and reviewers on request, and will be restored in the camera-ready version. The archive contains the optimisation-space definition, preprocessing and surrogate code, the candidate-production and retrieval implementations, the twelve-query suite, the parser prompts and the 40-utterance labelled parser-evaluation set, per-candidate confirmation results, the analysis notebooks, seeds, and a dependency list. Trial-level configuration and metric data are released for both use cases; only the proprietary Proditec images are withheld. The dataset-specific all-method metadata files also preserve each primary factual row identifier.

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: